CSCI 5541 NLP HW5

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1 Introduction

The goal of this assignment is to study how prompting affects the behavior of small-sized large language models. About Task family selected, I selected Task 3C: multiple-choice commonsense reasoning. I investigate whether different prompting methods can improve the performance of small language models on commonsense multiple choice, especially in terms of accuracy and producing valid answer formats.

2 Related Work

2.1 Model Selection and Justification

In this project, I selected two small open-weight instruction-tuned language models: Qwen2.5-3B-Instruct and SmolLM3-3B(using hyperlink here). Both are around the 3B-parameter scale, which makes them suitable for comparing prompting methods under limited computational resources such as Google Colab.

The two models also differ in an important way. Qwen2.5-3B-Instruct behaves like a standard instruction-tuned chat model, while SmolLM3-3B supports an optional reasoning or "thinking" mode. For this project, I explicitly disabled SmolLM3's thinking mode so that both models would be evaluated under more comparable output conditions and would be less likely to produce extra reasoning text. This makes the comparison more consistent for constrained-output tasks, where the model is required to return only a short, machine-readable answer such as a single label from A to E in task 3.

In detail, SmolLM3-3B and Qwen2.5-3B-Instruct are both small decoder-in the 3B range, but they differ in architecture and training emphasis. SmolLM3-3B uses GQA and NoPE, and it was pretrained on 11.2T tokens with additional reasoning-focused post-training on 140B tokens, suggesting a stronger focus on reasoning behavior. In contrast, Qwen2.5-3B-Instruct uses a more standard instruction-tuned Transformer architecture with GQA and RoPE. Therefore, although the two models are similar in scale and are fair to compare under the same compute budget.

2.2 Prompt method

I compared three prompting methods: zero-shot prompting, plan-and-solve prompting, and self-refine prompting. First, I use Zero-shot prompting which without providing task-specific demonstrations. I consider this property could that is method as a simple baseline. Second, The Plan-and-solve prompting which introduce by Wang et al. (2023), the key idea is to ask the model to first form a plan by decomposing the problem into smaller steps and then solve the problem according to that plan. Third, Self-refine prompting which proposed by Madaan et al. (2023), ask model first produces an initial answer, then checks that answer, and finally refines it. These three methods were chosen because they represent increasingly structured forms of inference-time guidance: zero-shot prompting provides direct instruction only, plan-and-solve adds explicit decomposition, and self-refine adds a self-evaluation step before the final answer.

Following are the limitation of each method: zero-shot: suitable for simple classification or multiple-choice tasks, but weaker on reasoning. plan-and-solve: suitable for multi-step reasoning, but may sensitive to the expressions in prompts. self-refine: suitable when output quality and format checking matter, but may increase latency.

3 Experimental Details

For Task 3C, I used the CommonsenseQA dataset is a new multiple-choice question answering dataset that requires different types of commonsense knowledge to predict the correct answers . It contains 12,102 questions with one correct answer and

four false answers. And each question has five answer choices with exactly one correct answer.

```
Example 1:
Input:
The sanctions against the school were a punishing blow, and
      they seemed to what the efforts the school had made to
      change?
"A", "ignore";
"B", "enforce";
"C", "authoritarian";
"D", "yell at";
"E", "avoid"
Output: Answer: A

Example 2:
Input:
Johnny sat on a bench and relaxed after doing a lot of work
      on his hobby. Where is he?
"label":
"A", "state park";
"B", "bus depot";
"C", "garden";
"D", "gym";
"E", "rest area"
Output: Answer: C
```

Experiments were conducted on Google Colab using an NVIDIA T4 GPU with approximately 15 GB of VRAM and standard Colab RAM. The software environment used Python 3, PyTorch, Hugging Face Transformers, and the Datasets library.

3.1 Prompt Templates

The following prompt templates were used for the three prompting methods in Task 3C.

3.1.1 Zero-shot prompting

```
System: You are a careful assistant.

User: Choose the best answer to the multiple-choice question.

Question:
{question}

Choices:
A. {choice_A}
B. {choice_B}
C. {choice_C}
D. {choice_D}
E. {choice_E}

Output constraints:
- Answer with exactly one letter.
- Allowed outputs: A, B, C, D, or E.
- Do not explain.
- Output only the final answer.
```

3.1.2 Plan-and-solve prompting

```
System: You are a careful reasoning assistant.

User: Solve the task in two stages.

Stage 1: Briefly compare the answer choices against the
      question.
Stage 2: Decide which option best answers the question.

Question:
{question}

Choices:
A. {choice_A}
B. {choice_B}
C. {choice_C}
D. {choice_D}
E. {choice_E}

Output constraints:
- Exactly one letter
- Allowed outputs: A, B, C, D, or E
- No explanation
```

3.1.3 Self-refine prompting

```
System: You are a careful assistant that checks its own
      output.

User: Choose the best answer to the multiple-choice question.

Question:
{question}

Choices:
A. {choice_A}
B. {choice_B}
C. {choice_C}
D. {choice_D}
E. {choice_E}

Instructions:
1. Draft an answer internally.
2. Check whether the answer matches the question and choices.
3. Check whether the answer follows the output constraint.
4. Output only the final revised answer.

Output constraints:
- Exactly one letter
- Allowed outputs: A, B, C, D, or E
- No explanation
```

Evaluation Metrics: I used exact match and format validity. Exact match checks whether the model's answer is the same as the correct label, because Task 3C is a multiple-choice task with one correct answer. Format validity checks whether the output follows the required format, such as giving a valid answer choice a single letter from A to E. I consider these metrics are appropriate because they measure both accuracy and whether the output can be follow as specific format.

4 Results and Analysis

4.1 Task 1

Following are 5 example and output of 2 model of task 1:

```
test_examples = [
    {
        "task": "Logical Negation",
        "prompt": "Which of the following is NOT a botanical
              berry: banana, grape, or peach? Answer with one
              word.",
        "expected": "{peach}"
    },
    {
        "task": "Ambiguous Reference",
        "prompt": "The trophy didn't fit into the brown
              suitcase because it was too large. What was too
              large? Answer with one noun phrase only.",
        "expected": "{the trophy, trophy}"
    },
    {
        "task": "Constrained Output",
        "prompt": "Write these three words in alphabetical
              order, separated by commas, with no spaces:
              green, blue, red.",
        "expected": "{blue,green,red}"
    },
```

```
{
    "task": "Data Extraction",
    "prompt": "ROI 2021: 15%. ROI 2022: 12%. Give the
          2022 ROI as a decimal. Answer with numbers
          only.",
    "expected": "{0.12}"
},
{
    "task": "Counterfactual Reasoning",
    "prompt": "If all birds suddenly lost their wings,
          could they still fly naturally? Answer yes or
          no only.",
    "expected": "{No}"
}
```

Task	Expected	Qwen2.5-3B	SmolLM3-3B
1	peach	banana	peach
2	trophy / the trophy	trophy	The trophy was too large.
3	blue,green,red	green,blue,red	blue, green, red
4	0.12	0.12	Explanation instead of number only
5	No	No	No.

Table 1: Comparison of Qwen2.5-3B and SmolLM3-3B on five representative test prompts.

In table 1, Qwen output wrong answer in Task 1 and 3, both response wrong ans but give correct format. On the other hand Smol give wrong output in task 4 and barely satisfactory response in task 2. Based on above examples, SmolLM3-3B appears stronger in some reasoning-heavy cases, but Qwen2.5-3B is more reliable at following strict output constraints.

4.2 Task 2

Based on the Task 1 comparison, I selected SmolLM3-3B for Task 2 as a focused prompting analysis because it showed stronger reasoning-oriented behavior on the original examples. As I mention before, I use zero-shot, plan and solve and self-refine prompting. Result of task 2 is in table 2.

In task2, Zero-shot is simple and concise, but it is less reliable on tasks with Data Extraction. Plan-and-solve can improve reasoning, but it often adds extra text, which hurts performance badly when only the final answer is allowed. Self-refine performs best overall because it combines accuracy with better control of the final output format.

All in all, I will conclude that the output of Self-Refine is correct, zero shot as well except of data extraction. However, the response totally unacceptable using Plan and Solve method.

4.3 Task 3

I choose c in Task 3, method of prompt, dataset and example are already show in previous paragraph. Inspire by Geng et al., 2025, we should also consider the response time. Hence, I count in average time to generate each output token. The summary result is at table 3.

4.4 Qualitative Analysis

Across the 18 representative examples I basically choose the same question in different models and methods, list in appendix, the main pattern is that prompting helps only when it improves answer selection without breaking the required one-letter output format. Zero-shot was the most stable baseline for both models: it usually produced valid single-letter answers, but it sometimes failed on questions requiring stronger commonsense reasoning, such as the revolving-door and farmland examples(I think them choose a reasonable but not most accommodation answer). Qwen's self-refine output fixed the revolving-door example and produced the correct answer. In contrast, plan-and-solve showed mixed behavior: it remained format-valid for SmolLM3-3B but did not noticeably improve its reasoning, while for Qwen2.5-3B-Instruct it often hurt performance by generating intermediate reasoning text instead of a single valid label. These examples suggest that prompting helps most when it improves reasoning without increasing output complexity, and it hurts when the method encourages extra explanation that conflicts with strict structured-output constraints.

In terms of efficiency, the time per output token was fairly similar across most settings, generally around 0.06 seconds per token. In general, zero-shot is better than other methods. Specifically, Qwen2.5-3B-Instruct Plan-and-Solve have lowest time per token. That because this value is amortized by incorrect output format.

5 Conclusion

In this homework, I studied how different prompting methods affect the performance of small language models on constrained-output tasks, focusing on Task 3C with CommonsenseQA. I compared Qwen2.5-3B-Instruct and SmolLM3-3B under zero-shot, plan-and-solve, and self-refine prompting, and evaluated them using exact match, format validity, and time per output token. The results show that zero-shot and self-refine performed best overall, while plan-and-solve was not consistently helpful, especially under strict output constraints. In particular, Qwen2.5-3B-Instruct per-

Task	Zero-shot Output	Plan-and-Solve Output	Self-Refine Output
1	peach	"Stage 1: To answer correctly, I need to understand the botanical definition of a berry and then apply it to the given options. Stage 2: A botanical berry is a fleshy fruit"	peach
2	The trophy was too large.	Stage 1: To answer correctly, I need to identify the noun phrase that is too large, which is the only piece of information provided in the question. Stage 2: The question states,	trophy
3	blue,green,red	Stage 1: To answer correctly, I need to list the words "blue," "green," and "red" in alphabetical order, separated by commas, with no spaces. Stage 2:	blue, green, red
4	Explanation instead of number only	Stage 1: To answer correctly, I need to convert the 2022 ROI of 12% into a decimal. This involves dividing the percentage by 100. Stage 2: 12	0.12
5	No	Stage 1: To answer this question correctly, I need to consider the fundamental requirement for flight: the ability to generate lift through the movement of air over wings. If birds lose their wings, they	No

Table 2: Outputs of three prompting methods on five test examples using SmolLM3-3B with thinking mode disabled. I conclude that the output of Self-Refine is correct, zero shot as well except data extraction. However, the response totally unacceptable using Plan and Solve method.

Model	Prompting Method	Exact Match	Format Valid	Time / Output Token (s)
HuggingFaceTB/SmolLM3-3B	Plan-and-Solve	0.633	1.000	0.0650
HuggingFaceTB/SmolLM3-3B	Self-Refine	0.667	1.000	0.0669
HuggingFaceTB/SmolLM3-3B	Zero-shot	0.667	1.000	0.0641
Qwen/Qwen2.5-3B-Instruct	Plan-and-Solve	0.067	0.133	0.0553
Qwen/Qwen2.5-3B-Instruct	Self-Refine	0.700	1.000	0.0649
Qwen/Qwen2.5-3B-Instruct	Zero-shot	0.700	1.000	0.0583

Table 3: Task 3 results by model and prompting method, including average time per output token.

formed very poorly with plan-and-solve, whereas both models achieved strong and stable results with zero-shot and self-refine. The efficiency results were also fairly similar across most settings, suggesting that accuracy and output reliability were more important than speed in this task. Overall, these experiments suggest that for small language models, simple prompting methods are often sufficient for structured multiple-choice output tasks, while more complex prompting does not always improve performance and may even hurt it.

References

[1] Saibo Geng, Hudson Cooper, Michał Moskal, Samuel Jenkins, Julian Berman, Nathan Ranchin, Robert West, Eric Horvitz, and Harsha Nori. 2025. Jsonschemabench: A rigorous benchmark of structured outputs for language models.

[2] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-refine: Iterative refinement with self-feedback.

[3] Lei Wang, Wanyu Xu, Yihuai Lan, Zhiqiang Hu, Yunshi Lan, Roy Ka-Wei Lee, and Ee-Peng Lim. 2023. Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models.

A Appendix: Task 3 18 Qualitative Examples

```
Zero-Shot, HuggingFaceTB/SmolLM3-3B, commonsenseqa_val_000
Question: A revolving door is convenient for two direction
     travel, but it also serves as a security measure at a
     what?
Choices:
A. bank
B. library
```

```
C. department store
D. mall
E. new york
Expect: A
Output: C
Result: Incorrect

Zero-Shot, HuggingFaceTB/SmolLM3-3B, commonsenseqa_val_004
Question: James was looking for a good place to buy
      farmland. Where might he look?
Choices:
A. midwest
B. countryside
C. estate
D. farming areas
E. illinois
Expect: A
Output: D
Result: Incorrect

Zero-Shot, HuggingFaceTB/SmolLM3-3B, commonsenseqa_val_001
Question: What do people aim to do at work?
Choices:
A. complete job
B. learn from each other
C. kill animals
D. wear hats
E. talk to each other
Expect: A
Output: A
Result: Correct

Plan-And-Solve, HuggingFaceTB/SmolLM3-3B,
      commonsenseqa_val_000
Question: A revolving door is convenient for two direction
      travel, but it also serves as a security measure at a
      what?
Choices:
A. bank
B. library
C. department store
D. mall
E. new york
Expect: A
Output: C
Result: Incorrect

Plan-And-Solve, HuggingFaceTB/SmolLM3-3B,
      commonsenseqa_val_004
Question: James was looking for a good place to buy
      farmland. Where might he look?
Choices:
A. midwest
B. countryside
C. estate
D. farming areas
E. illinois
Expect: A
Output: D
Result: Incorrect

Plan-And-Solve, HuggingFaceTB/SmolLM3-3B,
      commonsenseqa_val_001
Question: What do people aim to do at work?
Choices:
A. complete job
B. learn from each other
C. kill animals
D. wear hats
E. talk to each other
Expect: A
Output: A
Result: Correct

Self-Refine, HuggingFaceTB/SmolLM3-3B, commonsenseqa_val_000
Question: A revolving door is convenient for two direction
      travel, but it also serves as a security measure at a
      what?
Choices:
A. bank
B. library
C. department store
D. mall
E. new york
Expect: A
Output: C
Result: Incorrect

Self-Refine, HuggingFaceTB/SmolLM3-3B, commonsenseqa_val_004
Question: James was looking for a good place to buy
      farmland. Where might he look?
Choices:
A. midwest
B. countryside
C. estate
D. farming areas
E. illinois
Expect: A
Output: D
Result: Incorrect

Self-Refine, HuggingFaceTB/SmolLM3-3B, commonsenseqa_val_001
Question: What do people aim to do at work?
Choices:
A. complete job
B. learn from each other
C. kill animals
D. wear hats
E. talk to each other
Expect: A
Output: A
Result: Correct

Zero-Shot, Qwen/Qwen2.5-3B-Instruct, commonsenseqa_val_000
Question: A revolving door is convenient for two direction
      travel, but it also serves as a security measure at a
      what?
Choices:
A. bank
B. library
C. department store
D. mall
E. new york
Expect: A
Output: D
Result: Incorrect

Zero-Shot, Qwen/Qwen2.5-3B-Instruct, commonsenseqa_val_004
Question: James was looking for a good place to buy
      farmland. Where might he look?
Choices:
A. midwest
B. countryside
C. estate
D. farming areas
E. illinois
Expect: A
Output: D
Result: Incorrect

Zero-Shot, Qwen/Qwen2.5-3B-Instruct, commonsenseqa_val_001
Question: What do people aim to do at work?
Choices:
A. complete job
B. learn from each other
C. kill animals
D. wear hats
E. talk to each other
Expect: A
Output: A
Result: Correct

Plan-And-Solve, Qwen/Qwen2.5-3B-Instruct,
      commonsenseqa_val_000
Question: A revolving door is convenient for two direction
      travel, but it also serves as a security measure at a
      what?
Choices:
A. bank
B. library
C. department store
D. mall
E. new york
Expect: A
Output: D
Result: Incorrect

Plan-And-Solve, Qwen/Qwen2.5-3B-Instruct,
      commonsenseqa_val_001
Question: What do people aim to do at work?
Choices:
A. complete job
B. learn from each other
C. kill animals
```

```
D. wear hats
E. talk to each other
Expect: A
Output: Stage 1: The question asks about what people
      generally aim to do at work. Looking at the options:

A. "complete job" - This suggests a goal of finishing tasks,
      which is common
Result: Incorrect

Plan-And-Solve, Qwen/Qwen2.5-3B-Instruct,
      commonsenseqa_val_002
Question: Where would you find magazines along side many
      other printed works?
Choices:
A. doctor
B. bookstore
C. market
D. train station
E. mortuary
Expect: B
Output: B
Result: Correct

Self-Refine, Qwen/Qwen2.5-3B-Instruct, commonsenseqa_val_004
Question: James was looking for a good place to buy
      farmland. Where might he look?
Choices:
A. midwest
B. countryside
C. estate
D. farming areas
E. illinois
Expect: A
Output: D
Result: Incorrect

Self-Refine, Qwen/Qwen2.5-3B-Instruct, commonsenseqa_val_006
Question: In what Spanish speaking North American country
      can you get a great cup of coffee?
Choices:
A. mildred's coffee shop
B. mexico
C. diner
D. kitchen
E. canteen
Expect: B
Output: A
Result: Incorrect

Self-Refine, Qwen/Qwen2.5-3B-Instruct, commonsenseqa_val_000
Question: A revolving door is convenient for two direction
      travel, but it also serves as a security measure at a
      what?
Choices:
A. bank
B. library
C. department store
D. mall
E. new york
Expect: A
Output: A
Result: Correct
```